

Date: 31/07/2026

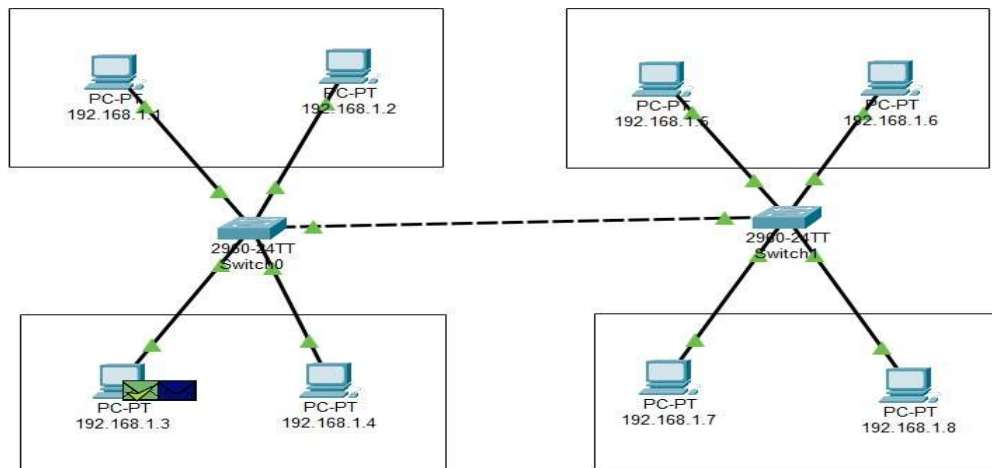
Lab Practical #05:

Study the concept of VLAN using packet tracer. **Practical**

Assignment #05:

1. Implement the different network structures in VLAN and VLAN trunking. Also check connectivity between them using ping command or PDU utility.

1st VLAN: -



```
192.168.1.3

Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:

Reply from 192.168.1.6: bytes=32 time=2ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```



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Switch0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/10

FastEthernet0/11

FastEthernet0/12

FastEthernet0/13

FastEthernet0/14

FastEthernet0/15

FastEthernet0/16

FastEthernet0/17

VLAN Configuration

VLAN Number: 20

VLAN Name: Faculty

VLAN No	VLAN Name
1	default
10	Student
20	Faculty
1002	fddi-default
1003	token-ring-default
1004	fddinet-default
1005	trnet-default

Equivalent IOS Commands

```
Switch(config)#interface FastEthernet0/7
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/7
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)# name Student
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name Faculty
Switch(config-vlan)#
```

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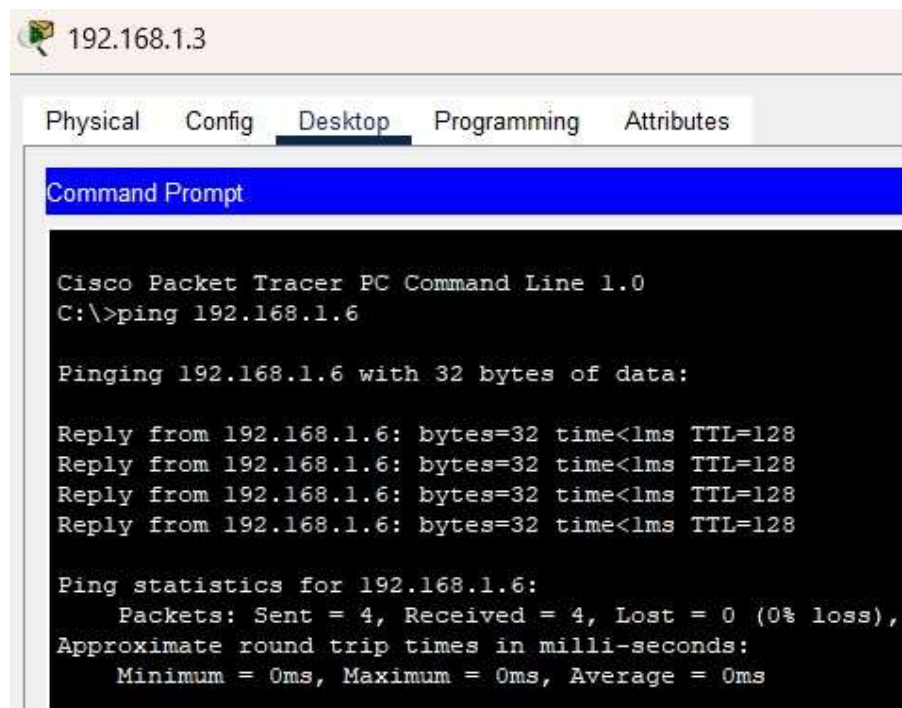
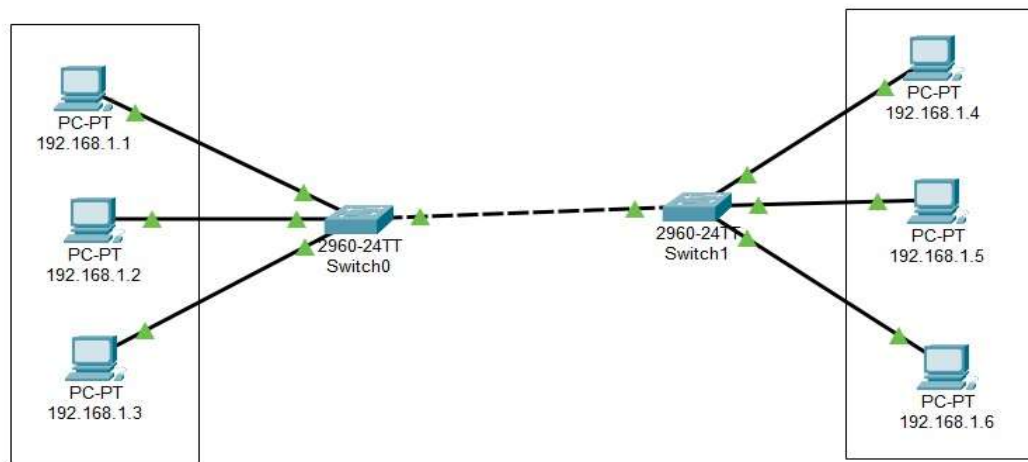


Steps :-

- Take 8 pc and assign Ip address and label to each pc.
- Take 2 Switch and Make 2 Lan with 4 pc each.
- Now there is local area network but We need to implement Virtual Lan database for implement Virtual Lan we need to configure Virtual Lan database in configure Section in switch.
- Make 2 New Virtual Lan configuration with unique Virtual Lan number and name.
- Now check the Port where the pc is connected and open that port configuration in switch and set Virtual Lan to access mode and select your Virtual Lan database name in dropdown.
- Now we need to connect both Switch but here we need to give Virtual Lan mode to trunk because there are number different Virtual Lan signal travel threw it.
- Now you configure two Virtual Lan in different local area network.

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2nd VLAN: -



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:

Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```



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Switch1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/10

FastEthernet0/11

FastEthernet0/12

FastEthernet0/13

FastEthernet0/14

FastEthernet0/15

FastEthernet0/16

FastEthernet0/17

VLAN Configuration

VLAN Number

20

VLAN Name

G2

Add

Remove

VLAN No	VLAN Name
1	default
10	G1
20	G2
1002	fddi-default
1003	token-ring-default
1004	fddinet-default
1005	trnet-default

Equivalent IOS Commands

```
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)# name G1
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name G2
Switch(config-vlan)#
```

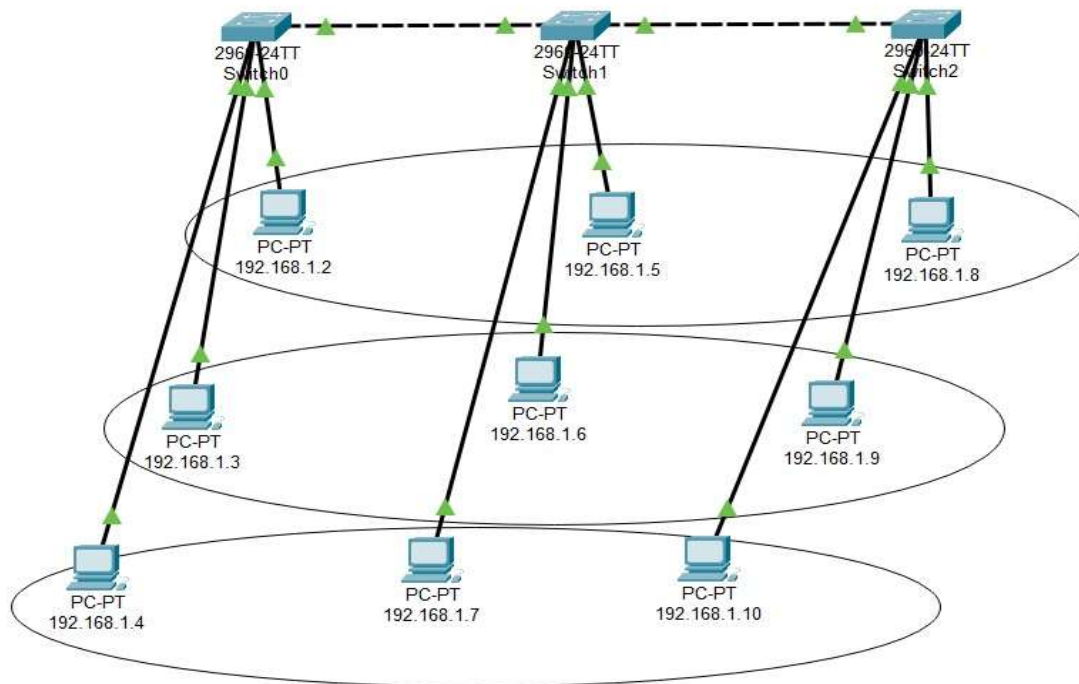
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Steps :-

- Take 6 pc and assign Ip address and label to each pc.
- Take 2 Switch and Make 2 Lan with 3 pc each.
- Now there is local area network but We need to implement Virtual Lan database for implement Virtual Lan we need to configure Virtual Lan database in configure Section in switch.
- Make 2 New Virtual Lan configuration with unique Virtual Lan number and name.
- Now check the Port where the pc is connected and open that port configuration in switch and set Virtual Lan to access mode and select your Virtual Lan database name in dropdown.
- Now we need to connect both Switch but here we need to give Virtual Lan mode to trunk because there are number different Virtual Lan signal travel threw it.
- Now you configure two Virtual Lan in different local area network.

3rd VLAN: -



```
192.168.1.4

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.9

Pinging 192.168.1.9 with 32 bytes of data:

Reply from 192.168.1.9: bytes=32 time<1ms TTL=128
Reply from 192.168.1.9: bytes=32 time=1ms TTL=128
Reply from 192.168.1.9: bytes=32 time<1ms TTL=128
Reply from 192.168.1.9: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



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Switch0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/10

FastEthernet0/11

FastEthernet0/12

FastEthernet0/13

FastEthernet0/14

FastEthernet0/15

FastEthernet0/16

FastEthernet0/17

VLAN Configuration

VLAN Number30

VLAN NameAdmin

AddRemove

VLAN No	VLAN Name
1	default
10	Student
20	Faculty
30	Admin
1002	fddi-default
1003	token-ring-default
1004	fddinet-default
1005	trnet-default

Equivalent IOS Commands

```
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)# name Student
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name Faculty
Switch(config-vlan)#vlan 30
Switch(config-vlan)# name Admin
Switch(config-vlan)#
```

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Steps:

- Take 9 PCs and assign IP address and label to each PC.
- Use 3 switches and make a LAN where each switch connects to 3 PCs.
- Now there is local area network but we need to implement Virtual LAN database.
- For implementing Virtual LAN, configure VLAN database in the Configure section of each switch.
- Make 3 new Virtual LANs with unique VLAN numbers and names (e.g., VLAN 10: Students, VLAN 20: Faculty, VLAN 30: Admin).
- Check the port where the PC is connected, open the Port Configuration in the switch, set port mode to Access, and select the correct VLAN from dropdown.
- Connect all switches together.
- On the ports between switches, set mode to Trunk because multiple VLAN signals will travel through it.
- Now you have 3 Virtual LANs configured across different LANs.
- Verify connectivity using PDU or ping. Devices in same VLAN should communicate successfully; different VLANs won't communicate without routing.